

# How Preeclampsia Is Predicted and Diagnosed — and Where BADIRA Fits

*Clinical definition, the two early-screening routes, and what each one requires*

**BADIRA** | بادرة

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## ABOUT BADIRA

**BADIRA** is a bilingual, mobile-first AI-based companion and a profile-guided, self-adaptive prediction framework designed to support early awareness of preeclampsia risk during pregnancy. It brings together information already known about a woman before the disease develops - such as gestational stage, medical and obstetric history, previous pregnancies, family history, current pregnancy characteristics, and available health measurements - to support a personalized assessment of future risk. **BADIRA** is designed for the stage before diagnosis, where identifying increased risk early can still support timely referral, closer monitoring, and clinician-directed preventive action.

*Know early. Act in time.*

## What each part of this report covers

This report brings together the clinical evidence needed to understand preeclampsia prediction for **BADIRA**. It moves from defining the disease and its outcome, through the main early risk-assessment routes and biomarker evidence, to their clinical requirements and local Saudi context.

| Part | Section                                          | What it covers                                                                                                                                                                                                                                         |
|------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Clinical definition of preeclampsia              | What preeclampsia is, how it is clinically diagnosed, what defines the later outcome (Y), and how prediction is separated from diagnosis.                                                                                                              |
| 2    | Route A — the risk-factor checklist              | History-based early risk assessment, the factors and combinations it uses, and the preventive management decision it can support.                                                                                                                      |
| 3    | Route B — FMF first-trimester combined screening | How maternal factors are combined with MAP, UtA-PI and PlGF to produce an individualized risk estimate, what the pathway requires, and its prediction performance.                                                                                     |
| 4    | The two routes side by side                      | How Route A and Route B differ in evidence, prediction approach, requirements, performance and availability. .                                                                                                                                         |
| 5    | Biomarkers: how research predicts preeclampsia   | Which biomarkers contribute to preeclampsia risk assessment and the important distinction between early prediction and later short-term clinical assessment.                                                                                           |
| 6    | Saudi clinical context                           | How preeclampsia risk assessment, prevention and diagnosis are applied in Saudi clinical practice and which resources and measurements are realistically available. <b>To be completed following the gynecologist consultation.</b>                    |
| -    | Conclusion                                       | What the clinical research establishes for <b>BADIRA StartSmart</b> : predict future preeclampsia from pre-disease information, interpret information in the woman's profile, and identify risk early enough for clinician-directed preventive action. |

## Part 1 — Clinical Definition of Preeclampsia

*Before anything can be predicted, the thing being predicted has to be defined precisely. This part sets out that definition.*

### 1.1 Preeclampsia in simple words

Preeclampsia is a pregnancy-related multisystem hypertensive disorder that usually develops after 20 weeks of pregnancy. It is more than “high blood pressure in pregnancy”: hypertension is accompanied by proteinuria and/or evidence of maternal organ involvement [1]. It can affect multiple maternal organs and systems, including the kidneys, liver, blood, brain, lungs, and blood vessels, and can also affect the placenta and baby. Placental dysfunction and maternal vascular dysfunction are important parts of the disease, although preeclampsia is complex and multifactorial. A woman can have clinically important preeclampsia without obvious symptoms, which is why routine antenatal monitoring is important [1].

**Key concept: High blood pressure is an important sign of preeclampsia, but it is not the whole disease.**

### 1.2 When can preeclampsia occur?

Preeclampsia typically develops after 20 weeks of pregnancy, and it is more common later in pregnancy. It can also develop after delivery, known as postpartum preeclampsia. Preeclampsia occurring before 20 weeks is very rare, rather than impossible, and should prompt investigation for unusual associated conditions [1].

Risk assessment can begin earlier in pregnancy, but it is important to distinguish risk of developing preeclampsia from an actual clinical diagnosis of preeclampsia.

### 1.3 Which organs and systems can be affected?

Because preeclampsia is a multisystem disorder, its effects are not limited to blood pressure.

| System          | What can happen                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| Blood vessels   | Dysfunction of the blood-vessel lining can contribute to increased vascular resistance and fluid leakage.      |
| Kidneys         | Kidney involvement can cause protein to leak into the urine and kidney function to decline.                    |
| Liver           | Liver injury can cause increased liver enzymes and may be associated with upper-abdominal pain.                |
| Blood           | Platelet levels can fall, and severe disease such as HELLP syndrome can involve breakdown of red blood cells.  |
| Brain           | Neurological involvement can cause headache or visual disturbances and, in severe cases, seizures (eclampsia). |
| Lungs           | Fluid can accumulate in the lungs (pulmonary edema), causing difficulty breathing.                             |
| Placenta / baby | Placental dysfunction can affect fetal growth and contribute to other pregnancy complications.                 |

[1]

*Important: These are possible effects of the disease. The specific organ findings used as clinical diagnostic criteria are covered separately in section 1.5.3.*

### 1.4 Why preeclampsia matters

Preeclampsia matters because it can progress quickly and unpredictably, sometimes even when the woman initially feels well.

As the condition progresses, it can cause serious maternal complications, including eclampsia (seizures), stroke, HELLP syndrome, kidney or liver injury, pulmonary edema, and placental abruption. It can also affect the pregnancy and baby through complications such as fetal growth restriction, medically indicated preterm delivery, and, in severe cases, fetal or maternal death [1].

Another important challenge is that management often involves timing. Definitive management ultimately requires delivery of the placenta. Other treatments, such as antihypertensive medication and magnesium sulfate, are used when appropriate to reduce complications and manage risk while clinicians determine the safest timing of delivery [1].

This can create a clinical balance between allowing more time for fetal maturity and avoiding increasing maternal or fetal risk if the condition worsens. For this reason, early prediction, regular monitoring, and timely clinical assessment are important.

Preeclampsia also has implications beyond the pregnancy, as a history of the condition is associated with increased long-term cardiovascular risk [1].

## 1.5 The clinical diagnosis

Preeclampsia is diagnosed using a combination of blood pressure measurements and evidence of proteinuria and/or maternal organ involvement. The basic clinical framework is: new-onset hypertension after 20 weeks + proteinuria OR qualifying maternal organ dysfunction [1]. Therefore, hypertension is essential to the standard diagnostic framework, but proteinuria is not always required.

### 1.5.1 Blood pressure criteria

New-onset hypertension means that blood pressure becomes abnormal after 20 weeks of pregnancy in a woman who was previously normotensive.

- Systolic blood pressure  $\geq 140$  mmHg — the upper number, representing arterial pressure when the heart contracts.
- Diastolic blood pressure  $\geq 90$  mmHg — the lower number, representing arterial pressure between heartbeats.
- Systolic  $\geq 140$  mmHg or diastolic  $\geq 90$  mmHg = hypertension.
- Systolic  $\geq 160$  mmHg or diastolic  $\geq 110$  mmHg = severe-range hypertension. [1]

Either value crossing its threshold is sufficient; both do not need to be elevated. Hypertension identified before 20 weeks is more consistent with chronic or pre-existing hypertension, although the woman's history and clinical context must be considered.

Ordinarily, an abnormal blood pressure is confirmed on two measurements at least 4 hours apart [1]. This is important because blood pressure can temporarily change because of factors such as anxiety, pain, physical activity, bladder fullness, incorrect cuff size, or white-coat effects.

An important ACOG distinction: if a woman has gestational hypertension and develops severe-range blood pressure, she is diagnosed and managed as having preeclampsia with severe features, even without proteinuria or another severe feature [1].

Severe-range blood pressure is particularly important because very high blood pressure creates an immediate maternal risk, including increased risk of stroke. Unlike ordinary hypertension, it can be confirmed within a short interval so that treatment is not delayed [1].

Blood pressure is one of the most important routinely measurable signs used to identify and monitor hypertensive disorders during pregnancy. It can help detect abnormal changes even when a woman has no obvious symptoms. When blood pressure reaches the severe range, it is also a direct maternal danger, particularly because of the increased risk of complications such as stroke [1].

Because high blood pressure in pregnancy has several different causes, and they are not the same illness:

- A woman can have had high blood pressure before she was pregnant (chronic hypertension).
- She can develop high blood pressure during pregnancy with nothing else wrong (gestational hypertension).
- Or she can develop high blood pressure plus evidence that her organs are being affected — that is preeclampsia.

**Two women can have identical blood pressure readings and have completely different conditions with completely different risks.**

### 1.5.2 Proteinuria criteria

Proteinuria means protein is present in the urine. In preeclampsia, it reflects kidney involvement, as damage to the kidney's filtering system allows protein to leak into the urine.

| Method                            | Diagnostic threshold                         |
|-----------------------------------|----------------------------------------------|
| 24-hour urine collection          | ≥300 mg per 24 hours                         |
| Protein-to-creatinine ratio (PCR) | ≥0.3                                         |
| Urine dipstick                    | 2+ when quantitative methods are unavailable |

[1]

### 1.5.3 Preeclampsia without proteinuria

A major point in the modern definition is that proteinuria is not required in every case of preeclampsia [1].

A woman with new-onset hypertension after 20 weeks can meet criteria for preeclampsia without proteinuria when qualifying new maternal organ dysfunction is present.

Qualifying findings include:

- Thrombocytopenia: platelet count <100,000/μL. Falling platelets indicate involvement of the blood/clotting system.
- Renal insufficiency: serum creatinine >1.1 mg/dL or doubling of the woman's baseline in the absence of other renal disease. The "doubling of baseline" criterion is important because a clinically meaningful decline in kidney function may occur even while the absolute creatinine value remains below 1.1 mg/dL.
- Impaired liver function: liver transaminases ≥2× the upper limit of normal and/or severe persistent right-upper-quadrant or epigastric pain that is not otherwise explained.
- Pulmonary edema: fluid accumulation in the lungs that can impair breathing and oxygenation.
- Neurological or visual features: new-onset headache that is unresponsive to medication and not explained by another diagnosis, or visual symptoms not explained by another diagnosis [1].

This is important because clinically significant preeclampsia can occur with little or no proteinuria.

### 1.5.4 The diagnostic logic

1. Is there new-onset hypertension after 20 weeks?

2. If yes, is proteinuria present?
3. If proteinuria is absent, is qualifying maternal organ dysfunction present?

If the required accompanying findings are present, the clinical framework for preeclampsia can be met [1].

**Key principle: Preeclampsia is not defined by blood pressure alone, and it is not defined by proteinuria alone. The diagnosis depends on the combination of hypertension with the appropriate accompanying clinical findings.**

### 1.5.5 From clinical diagnosis to the model outcome

For BADIRA, the clinical definition of preeclampsia also defines the outcome that the prediction model is trying to predict. Information collected before preeclampsia develops can be used as model inputs, while the outcome is determined later based on whether the woman actually develops clinically defined preeclampsia.

This creates an important separation:

#### **Before preeclampsia develops → Model inputs (X)**

Information already known about the woman, such as her characteristics, medical and pregnancy history, risk background, and available early clinical measurements.

#### **After follow-up → Outcome (Y)**

Whether the woman later developed clinically defined preeclampsia.

A symptom, individual risk factor, screening result, or AI risk score should therefore not be treated as the outcome itself. The outcome label should represent the actual clinical outcome, defined consistently according to the selected diagnostic guideline [1].

This distinction is important because BADIRA is designed to predict the future development of preeclampsia from information available before it occurs, rather than reproduce an existing risk score or diagnose the disease after it has developed.

## 1.6 Distinguishing preeclampsia from other hypertensive conditions

| Condition                                | Key distinction                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Chronic hypertension</b>              | Hypertension present before pregnancy or before 20 weeks                                      |
| <b>Gestational hypertension</b>          | New hypertension after 20 weeks without proteinuria or qualifying organ dysfunction           |
| <b>Preeclampsia</b>                      | New hypertension after 20 weeks with proteinuria and/or qualifying organ dysfunction          |
| <b>Preeclampsia with severe features</b> | Preeclampsia with severe-range blood pressure and/or specified severe maternal organ findings |

[1]

Some of the maternal organ abnormalities used to diagnose preeclampsia in the absence of proteinuria also overlap with the criteria used to classify preeclampsia as having severe features. These concepts therefore should not be treated as separate diseases [1].

For dataset labelling, preeclampsia and preeclampsia with severe features should not be treated as interchangeable labels. Preeclampsia represents the primary disease outcome, while severe features indicate a clinically important severity classification within confirmed preeclampsia. Therefore, BADIRA should define the primary preeclampsia label separately from any severity label, and the exact criteria used for each label should be documented consistently with the selected clinical guideline.

These disorders can share the same finding — high blood pressure — but they are not the same condition. Correct classification depends on when the hypertension began, the woman's previous history and baseline, and whether additional features develop.

**Key concept: A single blood-pressure reading does not tell the whole story. Pregnancy timing, previous measurements, and new clinical changes help determine what that reading means.**

### 1.6.1 Superimposed preeclampsia

Superimposed preeclampsia occurs when a woman with chronic hypertension develops preeclampsia during pregnancy. It can be harder to recognize because her blood pressure may already be elevated and, in some cases, proteinuria may already exist [1].

Clinicians therefore need to consider changes from the woman's own baseline, such as worsening hypertension or new clinical abnormalities [1].

This is an important example of why previous measurements matter: the meaning of a current value can depend on what was present earlier in pregnancy.

This illustrates the principle behind profile-guided interpretation: a current measurement cannot always be interpreted correctly without knowing the woman's previous history, baseline measurements, pregnancy timing, and clinical context.

## 1.7 Symptoms, warning signs, and what does NOT diagnose preeclampsia

Preeclampsia may also be present without obvious symptoms. Therefore, the absence of symptoms does not exclude the disease. Likewise, symptoms alone cannot confirm preeclampsia because many are non-specific and may occur during pregnancy for unrelated reasons. Diagnosis depends on the defined clinical criteria and clinical assessment described in Section 1.5 [1].

### 1.7.1 Symptoms that may occur

- Persistent or severe headache, particularly when it does not improve with usual treatment.
- Visual disturbance, such as blurred vision, flashing lights, spots, or temporary changes in vision.
- Pain in the right upper abdomen or epigastric area.
- New or unusual nausea and vomiting later in pregnancy.
- Sudden or marked swelling, particularly of the face or hands.
- Breathlessness or difficulty breathing.
- Reduced urine output.

[1]

*Important: These symptoms should be taken seriously when clinically concerning, but none of them alone proves that a woman has preeclampsia.*

### 1.7.2 What does NOT diagnose preeclampsia

The following findings may be clinically useful, may increase suspicion, or may prompt further assessment, but none should be treated as a diagnosis of preeclampsia by itself.

| Finding                         | Why it is not a diagnosis by itself                                                                                                   | What it represents                                                      |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| One high blood-pressure reading | A single non-severe reading requires appropriate confirmation and interpretation rather than being treated as a diagnosis on its own. | A clinical measurement that may prompt repeat measurement or assessment |
| Headache                        | Headache is common and can have many causes. The relevant neurological criterion is more specific than simply reporting a headache.   | A symptom requiring clinical interpretation                             |
| Swelling (oedema)               | Swelling is common during pregnancy and is not, by itself, a diagnostic criterion for preeclampsia.                                   | A non-specific finding                                                  |
| Nausea                          | Nausea can occur for many reasons during pregnancy and is not diagnostic on its own.                                                  | A non-specific symptom                                                  |
| A risk factor or family history | Risk factors change the probability of developing preeclampsia; they do not establish that the disease is currently present.          | An input to risk assessment                                             |
| An AI risk score                | A risk score estimates probability; it does not satisfy the clinical diagnostic criteria for preeclampsia.                            | A prediction or risk estimate                                           |
| First-trimester risk screening  | Early screening estimates future risk and may guide prevention or surveillance; it is not a diagnosis of current preeclampsia.        | Screening / risk estimation                                             |

[1]

**Key principle: A symptom, risk factor, screening result, single measurement, or AI output can contribute useful information, but it must not be presented as equivalent to a clinical diagnosis.**

Figure 1

How preeclampsia is found — and where Badira sits

Two ways to find a woman at risk before she is sick. One way to confirm she is sick. Each needs different things to be available.

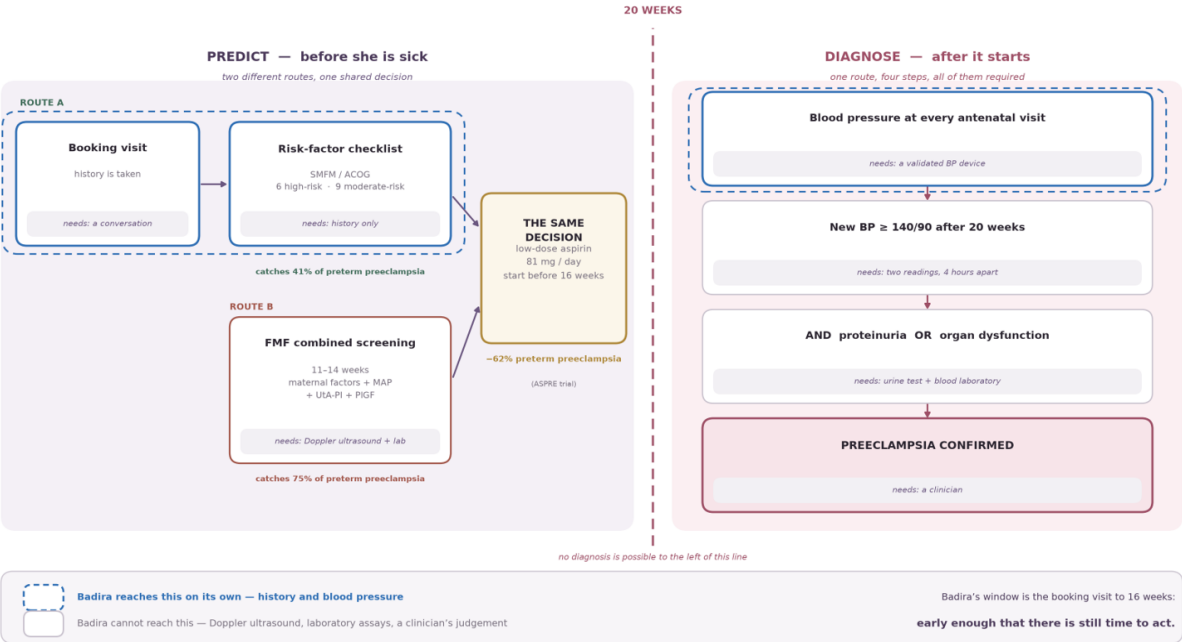


Figure 1. Preeclampsia clinical pathway: early risk prediction and prevention before disease onset versus later clinical assessment and diagnosis.

## Part 2 — Route A: the risk-factor checklist

*This is the first of the two prediction routes. It is the one that runs everywhere, because it needs nothing that a clinic might not have.*

### 2.1 What this screening route is

Before preeclampsia develops, clinicians can identify women who have characteristics associated with a higher risk of developing the condition. The SMFM/ACOG risk-factor approach uses information that is already known early in pregnancy, mainly maternal medical history, obstetric history, family history, current pregnancy characteristics, and basic maternal characteristics [4][5][6].

Unlike diagnosis, this route does not determine whether a woman currently has preeclampsia. It uses information available before the outcome occurs to identify increased future risk. These same types of pre-disease variables are therefore relevant to BADIRA as potential model inputs.

**Key point: this is risk assessment, not diagnosis. It asks who is more likely to develop preeclampsia later.**

### 2.2 High-risk factors

High-risk factors identify women with clinical or obstetric characteristics strongly associated with an increased risk of preeclampsia. Under the ACOG/SMFM risk-factor approach, the presence of at least one high-risk factor is sufficient for low-dose aspirin to be recommended when clinically appropriate [5].

| High-risk factor          | Meaning                                                               |
|---------------------------|-----------------------------------------------------------------------|
| Previous preeclampsia     | Preeclampsia in a previous pregnancy                                  |
| Multifetal gestation      | Current pregnancy with twins or more                                  |
| Chronic hypertension      | Pre-existing chronic high blood pressure                              |
| Type 1 or type 2 diabetes | Pregestational diabetes                                               |
| Renal disease             | Pre-existing kidney disease                                           |
| Autoimmune disease        | For example systemic lupus erythematosus or antiphospholipid syndrome |

[5][6]

**Decision rule: If a woman has at least one high-risk factor, ACOG/SMFM recommend low-dose aspirin to help prevent preeclampsia, when appropriate [5][6].**

### 2.3 Moderate-risk factors

Moderate-risk factors are characteristics associated with increased preeclampsia risk but that, individually, generally do not trigger the same preventive recommendation as a high-risk factor. The decision therefore considers whether multiple moderate-risk factors are present together [5][6].

| Moderate-risk factor | Meaning           |
|----------------------|-------------------|
| Nulliparity          | No previous birth |

| Moderate-risk factor                | Meaning                                                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Obesity                             | Prepregnancy BMI $\geq 30$ kg/m <sup>2</sup>                                                                                                     |
| Family history of preeclampsia      | Mother or sister had preeclampsia                                                                                                                |
| Maternal age $\geq 35$ years        | Age 35 years or older                                                                                                                            |
| Prior low-birthweight / SGA history | Relevant previous low-birthweight or small-for-gestational-age history                                                                           |
| Previous adverse pregnancy outcome  | Relevant adverse outcome in a previous pregnancy                                                                                                 |
| Long interpregnancy interval        | More than 10 years since the previous pregnancy                                                                                                  |
| IVF pregnancy                       | Current pregnancy conceived through in-vitro fertilization                                                                                       |
| Sociodemographic / contextual risk  | Social and healthcare conditions associated with inequity in preeclampsia risk; these should not be interpreted as biological racial differences |

[5][6]

**Decision rule: with more than one moderate-risk factor, low-dose aspirin should be considered/recommended according to the applicable ACOG/SMFM guidance** [5][6].

## 2.4 How the factors combine

Women often have more than one risk factor at the same time. In the checklist approach, these factors are handled using categorical decision rules: one high-risk factor can be sufficient to trigger preventive management, while multiple moderate-risk factors are considered together [5][6].

However, the checklist decision does not represent the woman's exact probability of developing preeclampsia. Different combinations can lead to the same checklist recommendation even though the women's overall profiles are different. A multivariable prediction model can preserve this additional information by considering several characteristics together rather than reducing the woman's profile to a single checklist category.

| Example profile                                      | High- risk factors | Moderate-risk factors | Checklist interpretation                                                                                   | What profile-guided prediction could preserve                                                                            |
|------------------------------------------------------|--------------------|-----------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Chronic hypertension only                            | 1                  | 0                     | One high-risk factor meets the checklist threshold                                                         | Her BP measurements, baseline history, and other available characteristics can still influence her estimated risk        |
| Twin pregnancy + obesity                             | 1                  | 1                     | Multifetal gestation already meets the high-risk threshold; obesity does not change the checklist decision | Both characteristics remain part of her individual profile rather than obesity being lost after the threshold is reached |
| Previous PE + chronic hypertension + type 2 diabetes | 3                  | 0                     | The checklist recommendation is reached regardless of whether one or several high-risk factors are present | The model can retain all three factors when estimating this woman's individual risk                                      |
| First pregnancy + BMI $\geq 30$ + age $\geq 35$      | 0                  | 3                     | Multiple moderate-risk factors meet the checklist threshold                                                | The actual values and combination can be considered rather than only counting three boxes                                |
| First pregnancy + family history                     | 0                  | 2                     | Multiple moderate-risk factors meet the checklist threshold                                                | Both factors remain part of the woman's profile and can be interpreted alongside other available measurements            |
| First pregnancy only                                 | 0                  | 1                     | The checklist alone does not meet the multiple-moderate-factor threshold                                   | Other information in her profile may still affect her estimated risk                                                     |

| Example profile         | High- risk factors | Moderate-risk factors | Checklist interpretation            | What profile-guided prediction could preserve                                         |
|-------------------------|--------------------|-----------------------|-------------------------------------|---------------------------------------------------------------------------------------|
| No listed factor at all | 0                  | 0                     | The woman is not flagged by Route A | Other validated predictors, if available, may still contribute information about risk |

[5][6].

**Key point: Risk factors can coexist, but their relationship to preeclampsia is not simply “one factor + one factor = a fixed amount of risk.” The checklist uses categories and thresholds for clinical decision-making [5][6]. A profile-guided prediction model instead aims to estimate risk from the woman’s combined profile, while the importance and interaction of individual variables must be learned and validated from appropriate data.**

## 2.5 What happens after a woman is flagged?

Identifying increased risk is useful because it can lead to preventive action before preeclampsia develops. Under ACOG/SMFM guidance, low-dose aspirin 81 mg/day is recommended for women who meet the applicable risk-factor criteria. It should be initiated between 12 and 28 weeks of gestation, optimally before 16 weeks, and continued until delivery [5][6].

This timing is important for BADIRA. The purpose of early prediction is not simply to tell a woman that she is at increased risk, but to identify that risk while there is still an opportunity for timely clinical assessment and clinician-directed preventive management.

*Important: Being classified as high risk does not mean that the woman currently has preeclampsia or that she will definitely develop it. Risk assessment identifies an increased probability and supports appropriate clinical action; it is not a diagnosis [5][6].*

## 2.6 What does this route require?

Route A relies mainly on information that can be collected early in prenatal care through the woman’s history and basic characteristics. This includes:

- **Maternal characteristics:** age and prepregnancy BMI
- **Medical history:** chronic hypertension, pregestational diabetes, renal disease, and autoimmune disease
- **Obstetric history:** previous preeclampsia, parity, previous adverse pregnancy outcomes, and interpregnancy interval
- **Current pregnancy:** multifetal pregnancy and IVF pregnancy
- **Family history:** family history of preeclampsia [5][6].

**No specialized preeclampsia biomarker testing or uterine artery Doppler is required for the Route A checklist. These measurements belong to more advanced multivariable screening approaches discussed in Part 3.**

## Connection to BADIRA

Because these characteristics are known before preeclampsia develops, they can serve as candidate pre-event model inputs (X) for BADIRA. Importantly, BADIRA would preserve them as part of the woman’s individual profile rather than reducing the information immediately to a checklist result such as “high risk” or “not high risk.”

This is important for Profile-Guided prediction: the model should receive what is actually known about the woman before the outcome occurs, while the later clinically confirmed preeclampsia outcome remains **Y**, as defined in Part 1

## 2.7 Main limitation

The checklist identifies increased risk but cannot tell with certainty who will develop preeclampsia. Its result also depends on having a complete and accurate history. A woman without recognized checklist risk factors can still develop the disease [5][6].

## 2.8 The updated 2026 SMFM checklist

The 2026 SMFM update is reflected in the risk-factor tables above, including the addition of IVF pregnancy as a moderate-risk factor and updated treatment of sociodemographic risk factors [6].

The sociodemographic update is particularly important for BADIRA. Risk associated with race in U.S.-based guidance should not be interpreted as evidence of a biological difference between racial groups. Instead, such associations may reflect differences in social conditions, access to care, healthcare experiences, and other contextual factors [6].

Because BADIRA StartSmart is intended for use in Saudi Arabia, U.S.-specific demographic categories should therefore not be transferred automatically into the model as biological risk factors. Candidate variables should be clearly defined, clinically justified, and evaluated using data relevant to the population in which BADIRA will be used.

This is also relevant to BADIRA's Data Profiler: before a variable learned from one dataset or population is used for prediction in another, BADIRA should consider whether the training population and variable definition are appropriate for the woman being assessed.

## 2.9 Clinical path in one view

1. Initial prenatal visit
2. Collect medical, obstetric and family history + maternal characteristics
3. Check high-risk and moderate-risk factors
4.  $\geq 1$  high-risk factor  $\rightarrow$  low-dose aspirin recommended when appropriate
5.  $> 1$  moderate-risk factor  $\rightarrow$  low-dose aspirin considered/recommended according to the applicable guidance
6. Continue antenatal surveillance
7. Risk screening  $\neq$  diagnosis of preeclampsia

## Part 3 — Route B: FMF first-trimester combined screening

*This is the second prediction route, and the one usually described as the gold standard. It measures more, catches more, and needs more.*

### 3.1 What is the FMF screening model?

The Fetal Medicine Foundation (FMF) model estimates a woman's risk of developing preeclampsia, especially preterm preeclampsia, at about 11–14 weeks. It combines maternal characteristics and medical/obstetric history with measured biomarkers [7][8].

**Key point: The result is an individual risk estimate — not a diagnosis of preeclampsia.**

## 3.2 What does it measure?

The main high-performance FMF first-trimester combination is maternal factors + mean arterial pressure (MAP) + uterine artery pulsatility index (UtA-PI) + placental growth factor (PLGF) [9].

PAPP-A can also be included, especially when already available from first-trimester screening, but FMF evidence shows that it does not significantly improve screening when PLGF is already included [9].

### 3.2.1 Maternal factors

These establish the woman's baseline risk. Examples include age, weight and height, previous pregnancy/preeclampsia history, chronic hypertension, diabetes, method of conception, family history of preeclampsia, and relevant autoimmune disease [7][9].

**Needed: patient history and/or medical record. No special lab or imaging equipment.**

### 3.2.2 Mean arterial pressure (MAP)

MAP represents average arterial pressure during one cardiac cycle. In FMF screening, blood pressure is measured using a standardized protocol and incorporated into the risk model after appropriate adjustment [7][9].

**Needed: validated automated blood-pressure monitor, correct cuff size, and standardized measurement technique.**

### 3.2.3 Uterine artery Doppler (UtA-PI)

This assesses resistance to blood flow in the uterine arteries supplying the uterus and placenta. The FMF protocol measures the pulsatility index (PI) of the left and right uterine arteries and uses their mean [7][9].

**Needed: ultrasound machine with color and pulsed-wave Doppler + a trained/certified operator following the FMF protocol.**

### 3.2.4 Placental growth factor (PLGF)

PLGF is a protein involved in placental blood-vessel development. Its level in maternal blood provides information related to placental function and is used as a biochemical marker in the model [7][9].

**Needed: maternal blood sample + laboratory assay/platform that measures PLGF.**

### 3.2.5 PAPP-A

PAPP-A (pregnancy-associated plasma protein-A) is a placental protein measured in maternal blood and is often available from first-trimester aneuploidy screening [9].

**Needed: maternal blood sample + compatible laboratory assay.**

*Important: PAPP-A can be included, but the core high-performance FMF combination is maternal factors + MAP + UtA-PI + PLGF because adding PAPP-A when PLGF is already included does not significantly improve screening performance [9].*

## 3.3 How are the measurements combined?

The measurements are not interpreted as separate diagnoses. The model starts with prior risk from maternal factors and history and combines it with the measured biomarkers to calculate a patient-specific risk. Biomarker measurements are standardized as multiples of the median (MoM) with appropriate adjustments before entering the model [9].

**This is the sentence that separates Route B from Route A. Route A counts boxes. Route B multiplies a prior risk by measured evidence and returns one number for one woman.**

## 3.4 Why this matters for BADIRA

The FMF pathway shows that advanced screening needs more than information entered into an app. Maternal history can be collected digitally, but UtA-PI requires Doppler ultrasound and PlGF/PAPP-A require blood testing and laboratory assays [7][9].

### 3.5 Requirements at a glance

| Component               | Purpose                              | What is needed                                             |
|-------------------------|--------------------------------------|------------------------------------------------------------|
| <b>Maternal factors</b> | Baseline risk                        | History / medical record                                   |
| <b>MAP</b>              | Maternal blood pressure              | Validated BP device + correct cuff + standardized protocol |
| <b>UtA-PI</b>           | Uterine artery blood-flow resistance | Doppler ultrasound + trained operator                      |
| <b>PlGF</b>             | Placental biomarker                  | Blood sample + laboratory assay                            |
| <b>PAPP-A</b>           | Additional placental marker          | Blood sample + laboratory assay                            |

[7][9]

### 3.6 Prediction Performance: What Additional Measurements Add

The performance of first-trimester preeclampsia screening improves as information beyond maternal history is added. Importantly, detection rates should only be compared when they refer to the same clinical outcome and the same false-positive rate (FPR).

For preterm preeclampsia (<37 weeks) at a 10% false-positive rate, a large FMF competing-risks study reported:

| Screening approach                                      | Detection rate for preterm PE (<37 weeks) | False-positive rate |
|---------------------------------------------------------|-------------------------------------------|---------------------|
| <b>Maternal factors alone</b>                           | 49%                                       | 10%                 |
| <b>Maternal factors + MAP</b>                           | 60%                                       | 10%                 |
| <b>Full FMF: maternal factors + MAP + UtA-PI + PlGF</b> | 75%                                       | 10%                 |

[9][14]

A simpler intermediate screening approach combines maternal factors with mean arterial pressure (MAP). At a 10% false-positive rate, first-trimester screening using maternal factors plus MAP detected about 60% of preeclampsia requiring delivery before 37 weeks and about 65% of cases requiring delivery before 34 weeks [14]. This represents an improvement over maternal factors alone, while remaining less resource-intensive than full FMF screening because it requires standardized blood-pressure measurement but not uterine artery Doppler or placental biomarker testing. For BADIRA, this combination is particularly relevant because it shows how prediction performance may improve when an accessible clinical measurement such as blood pressure is added to the woman's profile without requiring the full FMF infrastructure.

For earlier-onset disease, first-trimester combined screening performs even better. Published studies have reported detection of approximately 88–90% of preeclampsia requiring delivery before 34 weeks at a 10% false-positive rate when maternal characteristics are combined with biophysical and biochemical markers [15]. This shows that combined screening is particularly effective for identifying women at risk of earlier and more clinically serious forms of preeclampsia.

The same study found that the full FMF combination detected 47% of term preeclampsia, compared with 38% using maternal factors alone, at the same 10% false-positive rate [9].

**Key point: Maternal history already contains predictive information, but prediction performance is higher when maternal factors are combined with standardized clinical and biomarker measurements. In the study above, the full FMF combination of maternal factors + MAP + UtA-PI + PlGF detected 75% of preterm preeclampsia compared with 49% using maternal factors alone, at the same 10% false-positive rate [9]. This demonstrates the additional predictive information available from combined clinical measurements while also highlighting the greater resources required for the full FMF pathway.**

### 3.7 Other models, and why FMF is the reference

FMF is not the only prediction model for preeclampsia. A systematic review and meta-analysis identified 52 externally validated prediction models, using different combinations of maternal characteristics, medical history, and biomarkers [13]. However, most alternative models had more limited external validation and variable performance across populations.

The FMF competing-risks model for preterm preeclampsia currently has the strongest external-validation evidence. It has been validated across numerous settings, and the full FMF model combining maternal factors with MAP, UtA-PI and PlGF showed strong discrimination and calibration for predicting preterm preeclampsia [13].

FMF is also clinically important because its first-trimester screening approach was used in the ASPRE trial to identify women at high risk for preterm preeclampsia for aspirin prophylaxis [10].

**Key point: There is no shortage of possible prediction models. What matters is whether a model is appropriate for the available inputs, population, and data quality, and whether its performance has been externally validated. This supports BADIRA's approach of considering the woman's available profile and the characteristics of the supporting data rather than assuming that one prediction pathway is equally suitable in every setting.**

### 3.8 FMF pathway in one view

1. 11–14 week visit
2. Collect maternal factors/history
3. Measure MAP
4. Measure UtA-PI by Doppler ultrasound
5. Measure PlGF (and PAPP-A if used/available)
6. Combine inputs in FMF competing-risks model
7. Output: patient-specific risk estimate
8. High-risk result → clinical prevention/follow-up; NOT a diagnosis

[7][9]

## Part 4 — The two routes side by side

### 4.1 Same decision, different evidence

|              | Route A — risk-factor checklist                                           | Route B — FMF combined screening                            |
|--------------|---------------------------------------------------------------------------|-------------------------------------------------------------|
| What it uses | Medical, obstetric and family history plus basic maternal characteristics | The same history, plus MAP, uterine artery Doppler and PlGF |

|                               | Route A — risk-factor checklist                                                                                 | Route B — FMF combined screening                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>How it decides</b>         | Counts factors against a fixed list and applies a rule                                                          | Multiplies a prior risk by measured evidence and returns one patient-specific number                           |
| <b>When</b>                   | Any time from the booking visit                                                                                 | A fixed window, 11–14 weeks                                                                                    |
| <b>What it needs</b>          | Maternal history and basic characteristics; no specialized biomarker testing or uterine artery Doppler required | A validated BP device, a Doppler ultrasound machine, a certified operator, and a PlGF laboratory assay         |
| <b>Prediction performance</b> | Lower predictive performance than full combined screening; performance depends on the checklist/guideline used  | Full FMF screening detects about 75% of preterm PE (<37 weeks) at a 10% false-positive rate                    |
| <b>What it outputs</b>        | A categorical risk/management decision based on whether guideline criteria are met                              | An individualized probability/risk estimate                                                                    |
| <b>Where it can run</b>       | Antenatal settings where maternal history and basic characteristics can be collected                            | Clinics with first-trimester ultrasound and access to a PlGF assay                                             |
| <b>What it leads to</b>       | Guideline-directed preventive management; aspirin is optimally initiated before 16 weeks when indicated         | Guideline-directed preventive management; aspirin is optimally initiated before 16 weeks when indicated [5][6] |

[5][6][7][9]

## 4.2 Why both routes exist

Both routes identify women at increased risk, but they use different evidence. Route A uses accessible history-based risk factors, while Route B adds biophysical and biochemical markers for a more individualized risk estimate.

Route B generally offers better predictive performance, while Route A is more accessible because it relies on information routinely collected during antenatal care and does not require uterine artery Doppler or biochemical screening [5][6][7][9]. Early risk identification is important because it can support timely preventive action [10].

## 4.3 What the difference actually is

**A checklist counts. A model computes.**

Section 2.4 showed this directly. A woman with one high-risk factor and a woman with three high-risk factors receive the same output from the checklist, because the rule is a threshold on a count. The FMF model gives them different numbers, because it treats each factor as evidence to be weighed rather than a box to be ticked.

This is also why a woman with no listed checklist risk factor will not be flagged by Route A alone. A multivariable model may still identify increased risk when additional validated predictors, such as MAP, UtA-PI and PlGF, are considered alongside her maternal profile [9].

## 4.4 The availability problem

Route B requires several resources to be available within the appropriate first-trimester screening window: standardized MAP measurement, uterine artery Doppler performed using the required protocol, access to PlGF testing, and the necessary trained clinical personnel [7][9].

Where these resources and an appropriate first-trimester screening pathway are available, full combined screening can provide substantially more individualized risk information than the history-based checklist alone.

Where the complete Route B pathway is not available, women may instead be assessed using history-based risk-factor screening or the screening resources available in their clinical setting. This creates a gap between a highly accessible but less individualized approach and a higher-performing combined model that requires specialized measurements.

**This gap is important for BADIRA:** it raises the question of how much reliable, individualized prediction can be achieved from the information that is actually available for a specific woman, rather than assuming that every woman has access to the same measurements.

## Part 5 — Biomarkers: how research predicts preeclampsia

*Parts 2 and 3 described what clinics do. This part describes what the underlying research measures, and why. It answers the question: are there biomarkers that detect the risk of developing preeclampsia?*

### 5.1 What a biomarker is here

A biomarker in this context is a measurable biological indicator, such as a protein in maternal blood, that provides information about biological processes associated with preeclampsia. Some biomarker changes can be detected before the clinical signs of preeclampsia appear, allowing them to contribute to prediction or risk assessment [11].

This is possible because biological and placental changes associated with preeclampsia can begin before the disease becomes clinically detectable. Placental and angiogenic biomarkers can therefore provide information about these underlying processes before hypertension and other diagnostic features appear [11].

**The short answer: yes, biomarkers can contribute to predicting preeclampsia risk. For early first-trimester prediction, PlGF is particularly important and is already incorporated into the FMF combined screening model described in Part 3. Other biomarkers, including sFlt-1 and the sFlt-1/PlGF ratio, are more useful later in pregnancy for different clinical questions. These biomarkers require biological samples and laboratory testing; they cannot be obtained from questionnaire data alone.**

### 5.2 The biomarkers used or studied

| Biomarker         | What it is                                                                    | What happens in preeclampsia                                                                                                                                                            | Status                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PlGF              | Placental growth factor — a protein that drives placental blood-vessel growth | Lower levels in early pregnancy are associated with increased risk of later preeclampsia, particularly preterm disease                                                                  | In routine clinical use. Part of the FMF combined screening model                                                                                                  |
| sFlt-1            | Soluble fms-like tyrosine kinase-1 — a protein that blocks the action of PlGF | Levels tend to increase closer to the clinical onset of preeclampsia; its major established clinical value is later in pregnancy rather than as a stand-alone first-trimester predictor | Used later in pregnancy, particularly in combination with PlGF for assessment of suspected preeclampsia                                                            |
| sFlt-1/PlGF ratio | The balance between the two, expressed as one number                          | The ratio increases as the angiogenic imbalance associated with preeclampsia becomes more pronounced                                                                                    | Clinically useful for short-term assessment of suspected preeclampsia later in pregnancy; it is not the first-trimester prediction route used by BADIRA StartSmart |

| Biomarker | What it is                                                 | What happens in preeclampsia                                                                                                                           | Status                                                                                                        |
|-----------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| PAPP-A    | Pregnancy-associated plasma protein-A, a placental protein | Lower first-trimester levels are associated with increased risk of preeclampsia, but PAPP-A adds limited screening value when PlGF is already included | Available from first-trimester screening but not essential to the core FMF combination when PlGF is available |

[9][11][12]

### 5.3 The two different jobs biomarkers do

This distinction matters, and it is often blurred. The biomarkers are not all doing the same thing.

|                                 | First trimester (11–14 weeks)                                                                            | Second and third trimester (20–34 weeks)                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>The question being asked</b> | Will this woman develop preeclampsia later in this pregnancy?                                            | In a woman with suspected preeclampsia, what is the likelihood of developing preeclampsia in the short term?                        |
| <b>Which markers work</b>       | PlGF is the main biomarker used in the FMF combined screening model; PAPP-A may also provide information | The sFlt-1/PlGF ratio                                                                                                               |
| <b>What it is used for</b>      | Identifying increased risk early enough to support clinician-directed preventive management              | Supporting short-term clinical assessment and helping rule out preeclampsia over the following week in women with suspected disease |
| <b>Typical performance</b>      | As part of the FMF combined screening model : about 75% of preterm preeclampsia at 10% FPR               | A ratio $\leq 38$ gives a negative predictive value of about 99% over the following week                                            |

[9][10][11][12]

The first-trimester column represents prediction in the sense used by BADIRA: estimating future risk before preeclampsia develops and while preventive action may still be possible. The later-pregnancy column represents short-term clinical assessment of women with suspected disease and therefore belongs to the later clinical pathway shown in Figure 1.

### 5.4 What none of them can do

No biomarker alone diagnoses preeclampsia. A low PlGF level in the first trimester does not mean that a woman currently has preeclampsia; instead, it can contribute information about her risk of developing the condition later in pregnancy [9][11]. Clinical diagnosis remains separate from prediction and is based on the diagnostic criteria described in Part 1.

The biomarkers in Table 5.2 also require biological sampling and laboratory analysis. They therefore depend on clinical and laboratory infrastructure rather than questionnaire data alone [9][11][12].

## Part 6 — Saudi Clinical Context

*This section summarizes what the current Saudi literature shows about preeclampsia risk, awareness, and antenatal timing, and identifies the aspects of local clinical practice that still need confirmation through gynecologist and patient interviews.*

### 6.1 What the Saudi literature shows

Preeclampsia is an important maternal-health problem in Saudi Arabia. A 2025 systematic review and meta-analysis of 21 Saudi studies reported a pooled prevalence of 4.80% and documented serious maternal and fetal outcomes, including HELLP syndrome, eclampsia, prematurity, fetal growth restriction, and stillbirth [16].

Saudi studies also show limited awareness of preeclampsia. A 2023 study of Saudi women of reproductive age found important gaps in knowledge of the condition and its warning signs [17].

Timing of antenatal care is also relevant. A study of 277 women in Abha found a mean booking time of 12.2 weeks, with 36% attending after 12 weeks [18]. This is important because FMF combined screening is performed at approximately 11–14 weeks [7][9], so late booking can reduce the available screening window.

## 6.2 What is still unclear in Saudi practice

The Saudi literature confirms the burden of preeclampsia, limited awareness, and possible delays in entering antenatal care [16][17][18]. However, it does not clearly establish how preeclampsia risk screening is implemented in routine Saudi practice.

Important unanswered questions include:

- whether formal risk-factor checklists are routinely used;
- whether standardized MAP is used for first-trimester screening;
- how available uterine artery Doppler and PlGF testing are;
- what aspirin dose and timing are commonly used;
- whether women start and continue aspirin when indicated;
- whether preeclampsia is usually detected before or after symptoms appear;
- whether previous pregnancy information and measurements remain accessible across different healthcare settings.

These points should therefore be confirmed through the gynecologist and patient interviews rather than assumed from international guidelines.

## 6.3 Why this matters for BADIRA

Parts 2–4 showed that Route A is more accessible because it mainly uses maternal history and basic characteristics [5][6], while full FMF screening provides more individualized prediction but requires MAP, uterine artery Doppler, PlGF testing, trained personnel, and the correct first-trimester timing [7][9].

The Saudi evidence suggests that access to early screening cannot be assumed to be the same for every woman [18]. For BADIRA, this supports a profile-guided approach that uses the information actually available for each woman rather than assuming that every woman has access to the full FMF pathway.

The exact position of BADIRA in the Saudi clinical pathway should be finalized after comparing the literature with the gynecologist and patient interviews.

**Key point: Saudi evidence confirms that preeclampsia is an important local problem, awareness remains limited, and some women enter care late enough to narrow the first-trimester screening window. However, the current literature does not fully describe how advanced screening and prevention are implemented in routine Saudi practice.**

## Conclusion

This report establishes the clinical foundation for BADIRA's work on preeclampsia prediction. Preeclampsia is a clinical outcome that develops later in pregnancy and must remain clearly separated from early risk prediction. Information known before the disease develops — including maternal characteristics, medical and obstetric

history, current pregnancy factors, family history, and available early clinical measurements — can instead be used as prediction inputs, while the later clinically confirmed development of preeclampsia forms the outcome being predicted.

The research identified two main early risk-assessment routes. The risk-factor checklist is highly accessible and uses information available from the woman's profile and history, but it reduces combinations of factors to categorical management rules. FMF combined screening provides a more individualized risk estimate by combining maternal factors with MAP, UtA-PI and PlGF, but requires standardized measurements, specialized equipment, trained personnel and laboratory testing. Biomarker research further shows that biological measurements can contribute additional predictive information, while later biomarkers such as the sFlt-1/PlGF ratio answer a different clinical question and should not be confused with early prediction.

Together, these findings support the profile-guided principle behind BADIRA: risk information should be interpreted in the context of the individual woman rather than as isolated measurements or checklist boxes. They also establish that the value of prediction depends on timing. BADIRA is therefore concerned with estimating preeclampsia risk before the disease develops and while there is still an opportunity for clinician-directed preventive action — not with replacing clinical diagnosis.

The Saudi evidence adds an important local context to this foundation. It confirms that preeclampsia is a meaningful maternal-health problem, that awareness remains limited, and that some women enter antenatal care late enough to narrow the first-trimester screening window. However, the literature does not fully establish how risk screening, MAP, uterine artery Doppler, PlGF testing, aspirin use, and longitudinal records are implemented in routine Saudi practice. These points will therefore be confirmed through the gynecologist and patient interviews before BADIRA's exact position in the local clinical pathway is finalized.

## Sources

### Clinical definition (Part 1)

1. American College of Obstetricians and Gynecologists. Gestational Hypertension and Preeclampsia. ACOG Practice Bulletin No. 222. *Obstetrics & Gynecology*. 2020;135(6):e237–e260. doi:10.1097/AOG.0000000000003891.
2. National Institute for Health and Care Excellence. *Hypertension in pregnancy: diagnosis and management*. NICE Guideline NG133. Published 2019; updated 2023.
3. Magee LA, Brown MA, Hall DR, et al. The 2021 International Society for the Study of Hypertension in Pregnancy classification, diagnosis & management recommendations for international practice. *Pregnancy Hypertension*. 2022;27:148–169.

### Risk-factor screening (Part 2)

4. Society for Maternal-Fetal Medicine. Special Statement: Checklists for preeclampsia risk-factor screening to guide recommendations for prophylactic low-dose aspirin. 2020.
5. American College of Obstetricians and Gynecologists; Society for Maternal-Fetal Medicine. Low-Dose Aspirin Use for the Prevention of Preeclampsia and Related Morbidity and Mortality. Practice Advisory. December 2021; reaffirmed October 2022.
6. Society for Maternal-Fetal Medicine; Gharvey J, Combs CA; SMFM Patient Safety and Quality Committee. Special Statement: Updated checklists for preeclampsia risk-factor screening to guide recommendations for prophylactic low-dose aspirin. 2026.

### FMF screening (Part 3)

7. Fetal Medicine Foundation. *Preeclampsia Screening — FMF Certification*.
8. Fetal Medicine Foundation. *Assessment of Risk for Preeclampsia*.

### Prediction, performance and biomarker evidence (Parts 3–5)

9. O’Gorman N, Wright D, Poon LC, Rolnik DL, Syngelaki A, Wright A, et al. Accuracy of competing-risks model in screening for pre-eclampsia by maternal factors and biomarkers at 11–13 weeks’ gestation. *Ultrasound in Obstetrics & Gynecology*. 2017;49(6):751–755. doi:10.1002/uog.17399.
10. Rolnik DL, Wright D, Poon LC, O’Gorman N, Syngelaki A, de Paco Matallana C, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. *New England Journal of Medicine*. 2017;377(7):613–622. doi:10.1056/NEJMoa1704559
11. Chaemsaihong P, Sahota DS, Poon LC. First trimester preeclampsia screening and prediction. *American Journal of Obstetrics and Gynecology*. 2022;226(2S):S1071–S1097.e2. doi:10.1016/j.ajog.2020.07.020.
12. Zeisler H, Llurba E, Chantraine F, Vatish M, Staff AC, Sennström M, et al. Predictive value of the sFlt-1:PlGF ratio in women with suspected preeclampsia. *New England Journal of Medicine*. 2016;374(1):13–22. doi:10.1056/NEJMoa1414838.
13. Tiruneh SA, Vu CT, Moran LJ, Callander EJ, Allotey J, Thangaratinam S, et al. Externally validated prediction models for pre-eclampsia: systematic review and meta-analysis. *Ultrasound in Obstetrics & Gynecology*. 2024;63(5):592–604. doi:10.1002/uog.27490.
14. O’Gorman N, Nicolaides KH, Poon LCY. The use of ultrasound and other markers for early detection of preeclampsia. *Women’s Health (Lond)*. 2016;12(2):199–207. doi:10.2217/whe.15.95.
15. Tan MY, Syngelaki A, Poon LC, Rolnik DL, O’Gorman N, Delgado JL, et al. Screening for pre-eclampsia by maternal factors and biomarkers at 11–13 weeks’ gestation. *Ultrasound in Obstetrics & Gynecology*. 2018;52(2):186–195. doi:10.1002/uog.19112.

### Saudi clinical context (Part 6)

16. Nujoom RR, Babader RB, Aloufi AA, et al. *Preeclampsia in Saudi Arabia: A Systematic Review and Meta-Analysis of Prevalence, Risk Factors, Awareness, and Outcomes*. Cureus. 2025;17(12):e98969. doi:10.7759/cureus.98969.
17. Radwan A, Al Naji M, Alyoubi N, et al. *Awareness and Knowledge of Pre-eclampsia Among Saudi Women of Reproductive Age*. Cureus. 2023;15(11):e49233. doi:10.7759/cureus.49233.
18. Siddiqui AF. *Late antenatal booking and its predictors among mothers attending primary health care centers in Abha, Saudi Arabia*. Rawal Medical Journal. 2016;41(1):72–76.